Etherium Blockchain

Covered in CT-03 Slide

A list of elements present in the Ethereum blockchain is presented here:

- Ether cryptocurrency/tokens
- Keys and addresses
- Accounts
- Transactions and messages
- The EVM
- The Ethereum blockchain
- Nodes and miners
- The Ethereum network
- Smart contracts and native contracts
- Wallets and client software

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❑ Keys and addresses

- Keys and addresses are used in the Ethereum blockchain to represent ownership and transfer ether.
- The keys used are made up of pairs of private and public parts.
- The private key is generated randomly and is kept secret, whereas the public key is derived from the private key.
- Addresses are derived from public keys and are 20-byte codes used to identify accounts.

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❑ Keys Genertion

The process of key generation and address derivation is as follows:

1. First, a private key is randomly chosen (a 256-bit positive integer) under the rules defined by the elliptic curve secp256k1 specification. A private key is basically a valid nonzero 256-bit integer randomly picked up from the range of the elliptic curve's field size (order). Ethereum client software makes use of the operating system's random number generator to generate this number. Once this integer is picked up, it is represented in hex format (64 hexadecimal characters) as shown here: b51928c22782e97cca95c490eb958b06fab7a70b9512c38c36974f47b954ffc4.
2. The public key is then derived from this private key using the Elliptic Curve Digital Signature Algorithm (ECDSA) recovery function. A public key is an x, y coordinate, a point on an elliptic curve that satisfies the elliptic curve equation.

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The public key is represented in Ethereum as a hexadecimal string of 130 characters:
3aa5b8eefd12bdc2d26f 1ae348e5f383480877bda6f9e1a47f6a4afb35cf998ab847 f1e3948b1173622dafc6b4ac198c97 b18fe1d79f90c9093ab2ff9ad99260. b51928c22782e97cca95c490eb958b06fab7a70b9512c38c36974f47b954ffc4.

1. The public key is then derived from this private key using the Elliptic Curve Digital Signature Algorithm (ECDSA) recovery function.
2. An address is a unique identifier on the Ethereum blockchain, which is derived from the public key by hashing it through the Keccak-256 hash function and keeping the last 20 bytes of the hash produced. An example is shown here: 0x77b4b5699827c5c49f73bd16fd5ce3d828c36f32.